

# Master Fraud Intelligence Document

## Identity/KYC Attack Families

### FAMILY 1: SYNTHETIC IDENTITY FRAUD

#### A. Identity

| Field                          | Value                                                                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------------|
| Attack ID                      | SIF-001                                                                                                     |
| Attack Family                  | Synthetic Identity Fraud                                                                                    |
| Attack Subtypes / Variants     | Credit-building / Bust-out; Mule account creation; Synthetic identity farming                               |
| Primary Lifecycle Stage        | Identity / KYC (Stage 1)                                                                                    |
| Secondary / Cross-stage Stages | Account creation (Stage 2); Payment initiation (Stage 5); Settlement (Stage 13); Cash-out / Mule (Stage 16) |
| Target / Actor                 | KYC/onboarding systems, credit bureaus, payment platforms                                                   |
| Attack Objective               | Create a credible fake identity to open accounts, obtain credit, or conduct transactions undetected         |

#### B. Fraud Understanding

##### What is the attack?

Synthetic identity fraud involves creating fictitious identities by combining real and fabricated information—e.g., a real Social Security number with a fake name and address. Mastercard describes synthetic identities as "complete with personal histories, a web presence, social posts, photos, audio

and videos of an invented person". These identities are "cultivated" over months to build a credit history before a "bust-out" attack.

## Traditional mechanism

Before GenAI, fraudsters created synthetic identities manually, combining real and fabricated information. This required access to breached data, patience for manual identity cultivation, and knowledge of credit system mechanics. The attack was labour-intensive and not easily scalable.

## GenAI transformation

GenAI enables **industrial-scale generation** of hyperrealistic synthetic personas with coherent digital footprints. Mastercard data from Q1 2025 shows synthetic identities surged **300%**. Sumsu reports synthetic identity document fraud rose over 300% year-over-year in the U.S..

## Why GenAI is load-bearing

If GenAI is removed, the attacker loses the ability to:

- Generate thousands of unique, convincing synthetic personas rapidly
- Create coherent digital footprints (social media presence, online history) for each identity
- Automate the cultivation process across multiple identities simultaneously
- Scale the attack from dozens to thousands of accounts

The attack transforms from a niche, labour-intensive operation into an **industrial-scale automated campaign**.

## Prerequisites

- Access to GenAI tools (GANs, diffusion models, LLMs)
- Breached PII datasets (real identity fragments)
- Understanding of KYC verification logic
- Patience for cultivation period (30–365 days)

## Attack flow

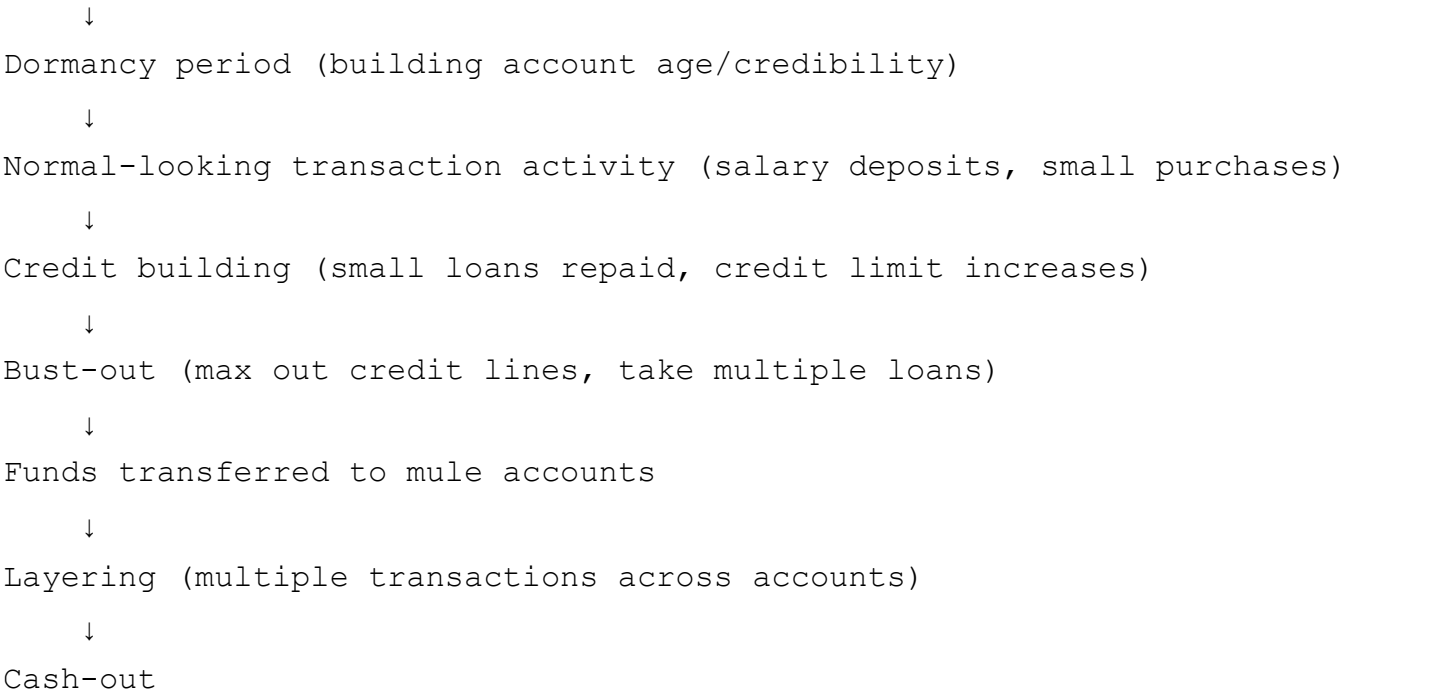
Synthetic identity creation (GenAI-generated persona + digital footprint)

↓

KYC/onboarding submission (documents, biometrics, V-CIP)

↓

Account creation (savings, credit card, digital wallet)



## Downstream payment consequence

The downstream impact includes credit losses, chargebacks, and significant AML exposure. Mastercard's 2025 payment fraud prevention report found that **61% of industry leaders** see synthetic identity fraud as the fastest-growing threat. Deloitte projects losses could reach **\$23 billion by 2030**.

## C. Detection Intelligence

| Signal Category | Specific Signals                                                          | Detection Method |
|-----------------|---------------------------------------------------------------------------|------------------|
| Identity/KYC    | Identity data inconsistencies (PII cross-check against CKYCR, DigiLocker) | Rule-based + ML  |
|                 | Digital footprint credibility (too new, inconsistent, or fabricated)      | ML               |
|                 | Document verification anomalies                                           | ML               |
|                 | Multiple verification attempts across identities                          | ML               |
| Device/Session  | Device emulation detection                                                | Rule-based       |
|                 | VPN/Proxy usage patterns                                                  | Rule-based + ML  |
|                 | Inconsistent device fingerprints across applications                      | ML               |
| Behavioral      | Unnatural typing/mouse patterns during onboarding                         | ML               |
|                 | Application completion time anomalies                                     | ML               |
|                 | Inconsistent application behaviour                                        | ML               |

| Signal Category      | Specific Signals                                                         | Detection Method             |
|----------------------|--------------------------------------------------------------------------|------------------------------|
| Transaction          | Too-rapid credit building (velocity anomalies)                           | ML                           |
|                      | Transaction patterns inconsistent with declared profile                  | ML                           |
|                      | Unusual transaction amounts/velocity                                     | ML                           |
|                      | Bust-out transaction sequences                                           | ML                           |
|                      | Funds movement to mule accounts                                          | ML                           |
| Network/Relationship | Shared phone numbers/addresses/email domains across "unrelated" accounts | ML (Graph analytics)         |
|                      | Connected to known fraud rings                                           | ML (Consortium intelligence) |
|                      | Entity resolution graph anomalies                                        | ML                           |
| Temporal             | Accounts with identical dormancy-to-activity patterns                    | ML                           |
|                      | Too-rapid credit building timeline                                       | ML                           |
|                      | Unusual transaction timing                                               | ML                           |

## Evasion considerations

An adaptive attacker could:

- Use unique phone numbers/addresses per synthetic identity (generated by GenAI)
- Create convincing social media histories and online presences for each identity
- Vary dormancy periods randomly rather than following predictable patterns
- Use device emulation and randomised fingerprints for each application
- Generate documents with GenAI that pass automated checks

## D. Simulation Intelligence

| Field                 | Value                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Simulation type       | Algorithmic                                                                                                                                   |
| Generation parameters | Identity data distributions; credit-building sequences; transaction pattern distributions; temporal patterns; network relationship generation |

| Field                           | Value                                                                                                                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthetic entities required     | Synthetic personas (name, DOB, address, occupation, income); Synthetic documents (ID cards, passports, utility bills); Digital footprint (email, phone, social media presence)                                      |
| Synthetic transactions required | Dormancy period (no transactions); Normal activity (salary deposits, bill payments, small purchases); Credit-building (loan applications, repayments, limit increases); Bust-out (large transactions, maxed credit) |
| Temporal behavior               | Gradual credit-building over 30–365 days; Normal-looking transaction patterns during cultivation; Sudden spike at bust-out                                                                                          |
| Network relationships           | Shared attributes across multiple synthetic identities; Connections to mule accounts; Fraud ring clustering                                                                                                         |
| Legitimate counterpart          | Thin-file customers (new immigrants, students); Legitimate credit-building customers; Users with limited digital history                                                                                            |
| Edge cases                      | Partially verified identities; Identities that pass initial KYC but fail later verification; Varying cultivation durations                                                                                          |

## E. Evidence

| Claim                                                                        | Source                                          | Confidence |
|------------------------------------------------------------------------------|-------------------------------------------------|------------|
| Synthetic identities surged 300% in Q1 2025                                  | Mastercard data (ItaliaOggi)                    | Verified   |
| Synthetic identity document fraud rose 300%+ in U.S. Q1 2025                 | Sumsb Q1 2025 Identity Fraud Trends             | Verified   |
| 61% of leaders see synthetic identity as fastest-growing threat              | Mastercard 2025 payment fraud prevention report | Verified   |
| Synthetic identities can pass initial checks and build credibility over time | PYMNTS/Mastercard/Trulioo report                | Verified   |
| Deloitte projects \$23B losses by 2030                                       | Deloitte Center for Financial Services          | Verified   |
| Open finance offers stronger identity verification                           | Mastercard                                      | Verified   |
| Mastercard Identity Network taps into 8B+ identity data points               | Mastercard                                      | Verified   |

## F. Product Mapping

| Component | Implementation |
|-----------|----------------|
|-----------|----------------|

| Component            | Implementation                                                                                                                                                                                                |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generate module      | Algorithmic generation of synthetic personas with consistent attributes; Transaction sequence generation for cultivation and bust-out phases; Network relationship generation for synthetic identity clusters |
| Defend module        | Entity resolution graph; Identity data cross-reference (CKYCR, DigiLocker); Digital footprint credibility scoring                                                                                             |
| Detection model      | Graph Neural Network for entity resolution; Anomaly detection for credit-building velocity; Behavioral biometrics for onboarding                                                                              |
| Potential mitigation | Real-time identity verification against trusted sources; Continuous trust scoring (not just event-based); Open finance data sharing for identity validation                                                   |
| Evaluation metric    | Precision (avoiding false positives on thin-file customers); Recall (catching synthetic identities before bust-out); AUC                                                                                      |

## FAMILY 2: GENAI DOCUMENT FORGERY

### A. Identity

| Field                          | Value                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| Attack ID                      | GDF-001                                                                                                    |
| Attack Family                  | GenAI Document Forgery                                                                                     |
| Attack Subtypes / Variants     | Passport forgery; Driver's license forgery; National ID forgery; Utility bill forgery                      |
| Primary Lifecycle Stage        | Identity / KYC (Stage 1)                                                                                   |
| Secondary / Cross-stage Stages | Account creation (Stage 2); Authentication (Stage 4)                                                       |
| Target / Actor                 | Document verification systems (OCR, document forensics, manual review)                                     |
| Attack Objective               | Submit fraudulent identity documents that pass verification, enabling account opening under false identity |

### B. Fraud Understanding

# What is the attack?

Generation of hyper-realistic forged identity documents using GenAI. Entrust's 2026 Identity Fraud Report notes that "what once required specialized software and design skills can now be achieved with an open-source model and a few prompts".

## Traditional mechanism

Before GenAI, fraudsters created forged documents using physical counterfeiting (printing fake IDs), Photoshop/manual image manipulation, or stolen blank documents. This required graphic design skills, access to printing equipment, and understanding of document security features. Physical counterfeits accounted for **47%** of document fraud attempts.

## GenAI transformation

GenAI enables the creation of hyper-realistic replicas of identity documents that can fool automated KYC systems. Digital forgeries now comprise over a third (**35%**) of document fraud attempts.

## Why GenAI is load-bearing

If GenAI is removed, the attacker loses:

- Ability to generate photorealistic documents without specialised design skills
- Ability to create documents for any jurisdiction/issuer on demand
- Ability to rapidly iterate and improve forgeries based on detection feedback
- Ability to scale from dozens to thousands of forged documents

## Prerequisites

- Access to GenAI image generation tools
- Template of legitimate document
- Understanding of verification logic
- Identity details to populate document

## Attack flow

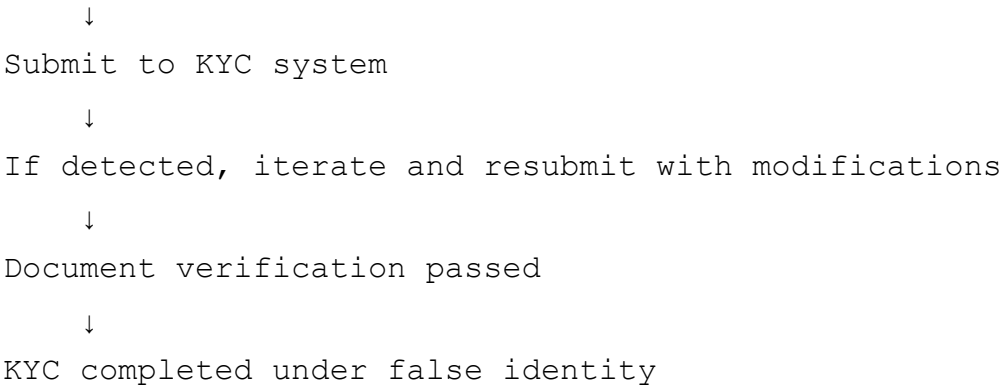
Obtain document template

↓

Generate document with desired details using GenAI

↓

Apply aging/noise effects to appear legitimate



## Downstream payment consequence

The forged document is the **enabler**—it provides the documented identity evidence needed to pass KYC. Once passed, all downstream payment fraud becomes possible under a verified identity that the institution believes is legitimate. This includes credit abuse, money movement, mule activity, and account takeover.

## C. Detection Intelligence

| Signal Category | Specific Signals                                                           | Detection Method                 |
|-----------------|----------------------------------------------------------------------------|----------------------------------|
| Identity/KYC    | Document pixel-level anomalies (compression artifacts, unnatural patterns) | ML (AI-driven forgery detection) |
|                 | Document metadata anomalies (creation date, software used)                 | Rule-based + ML                  |
|                 | Missing or inconsistent security features                                  | ML                               |
|                 | Inconsistent fonts, layout, or formatting                                  | ML                               |
|                 | Unnatural aging/wear patterns                                              | ML                               |
|                 | OCR data inconsistencies                                                   | Rule-based                       |
| Device/Session  | Multiple document submissions from same device                             | Rule-based                       |
|                 | Rapid resubmission after rejection                                         | Rule-based                       |
|                 | Unusual submission times                                                   | Rule-based                       |
| Behavioral      | Unnatural upload patterns                                                  | ML                               |
|                 | Documents submitted at unusual times                                       | Rule-based                       |
| Transaction     | (Post-account) transactions inconsistent with declared identity            | ML                               |



| Signal Category      | Specific Signals                                         | Detection Method             |
|----------------------|----------------------------------------------------------|------------------------------|
| Network/Relationship | Same document template used across multiple applications | ML (Document fingerprinting) |
|                      | Same GenAI artifact patterns across documents            | ML                           |
| Temporal             | Too-rapid iteration between submission and resubmission  | Rule-based                   |
|                      | Unusual verification timing                              | ML                           |

## Evasion considerations

An adaptive attacker could:

- Add realistic noise, compression artifacts, and aging effects to evade pixel-level detection
- Strip or forge metadata to appear authentic
- Introduce unique variations in each generated document to evade template detection
- Train GenAI models on security feature patterns
- Ensure extracted data matches claimed identity

## D. Simulation Intelligence

| Field                           | Value                                                                                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simulation type                 | Algorithmic                                                                                                                                                                        |
| Generation parameters           | Document type distribution; Forgery quality levels; Artifact patterns; Metadata generation                                                                                         |
| Synthetic entities required     | Synthetic document images (passports, DL, Aadhaar, utility bills); Documents with varying quality levels (pristine, aged, scanned, photographed); Metadata for synthetic documents |
| Synthetic transactions required | Document submission sequences (initial submission, rejection, resubmission)                                                                                                        |
| Temporal behavior               | Submission timing (off-hours weighted); Retry timing patterns                                                                                                                      |
| Network relationships           | Document template reuse across applications; Same GenAI artifact patterns across documents                                                                                         |
| Legitimate counterpart          | Poor-quality scans of legitimate documents; Damaged or worn legitimate documents; Documents from unfamiliar jurisdictions                                                          |

| Field      | Value                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------|
| Edge cases | Partially legible documents; Documents with unusual but legitimate features; Documents from multiple jurisdictions |

## E. Evidence

| Claim                                                                                   | Source                             | Confidence |
|-----------------------------------------------------------------------------------------|------------------------------------|------------|
| Digital forgeries comprise 35% of document fraud attempts                               | Entrust 2026 Identity Fraud Report | Verified   |
| Physical counterfeits account for 47% of document fraud attempts                        | Entrust 2026 Identity Fraud Report | Verified   |
| "What once required specialized software can now be achieved with an open-source model" | Entrust 2026 Identity Fraud Report | Verified   |
| National ID cards accounted for 46% of fraudulent document submissions                  | Entrust 2026 Identity Fraud Report | Verified   |

## F. Product Mapping

| Component            | Implementation                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Generate module      | Algorithmic generation of synthetic documents with varying quality levels and artifact patterns; Document metadata generation |
| Defend module        | AI-driven document forensics; Metadata analysis; Document fingerprinting                                                      |
| Detection model      | CNN-based forgery detection; Anomaly detection for document artifacts; OCR consistency checking                               |
| Potential mitigation | Multi-layered document verification; Real-time forgery detection; Document fingerprint database                               |
| Evaluation metric    | True Positive Rate (catching forgeries); False Positive Rate (not rejecting legitimate documents); Speed                      |

# FAMILY 3: DEEPFAKE INJECTION / IMPERSONATION

# A. Identity

| Field                          | Value                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Attack ID                      | DII-001                                                                                                              |
| Attack Family                  | Deepfake Injection / Impersonation                                                                                   |
| Attack Subtypes / Variants     | Camera injection (virtual camera); Face-swap injection; Voice cloning; Synthetic biometric data (fingerprints, iris) |
| Primary Lifecycle Stage        | Identity / KYC (Stage 1)                                                                                             |
| Secondary / Cross-stage Stages | Authentication (Stage 4); Payment initiation (Stage 5)                                                               |
| Target / Actor                 | Biometric verification systems, liveness detection, V-CIP agents, human reviewers                                    |
| Attack Objective               | Bypass biometric verification and liveness checks to create/access accounts under false identity                     |

# B. Fraud Understanding

## What is the attack?

Use of GenAI-generated synthetic video, audio, or images to bypass biometric verification and liveness detection—either through camera injection or direct impersonation. The iProov/MITRE ATLAS case study demonstrates how attackers use "readily available face-swapping tools and virtual camera applications" to inject deepfakes into mobile KYC workflows.

## Traditional mechanism

Before GenAI, fraudsters used **presentation attacks**: holding a photo of the victim in front of the camera, using a 3D mask, or playing a pre-recorded video. These attacks were relatively easy to defeat with basic liveness detection (blink detection, 3D depth sensing).

## GenAI transformation

GenAI enables **injection attacks** where synthetic video is fed directly into the camera pipeline, bypassing the physical camera entirely. This is fundamentally different from presentation attacks. The

iProov case study "combines real-time face-swap deepfake generation with camera injection on Android, targeting mobile onboarding and account access processes".

**Deepfakes now account for one in five (20%) biometric fraud attempts.** Instances of deepfaked selfies increased by **58% in 2025**. Injection attacks increased **40% year-over-year**.

Deepfakes are especially prevalent in financial services: crypto (60%), digital-first banks (22%), and payments/merchants (13%).

## Why GenAI is load-bearing

If GenAI is removed, the attacker loses:

- Ability to generate real-time face-swap deepfakes
- Ability to bypass liveness detection with synthetic video
- Ability to perform "defeat the sensor" attacks

The attack transforms from physical spoofing (photo, mask) to **digital injection** that bypasses the camera entirely—a qualitatively new attack vector.

## Prerequisites

- Access to victim photos/videos
- GenAI face-swap/deepfake tool
- Camera injection capability (virtual camera application)
- Knowledge of verification flow

## Attack flow

Obtain victim photos/videos (social media, data breaches)

↓

Generate real-time deepfake (face swap) using GenAI

↓

Set up camera injection (virtual camera application)

↓

Feed synthetic video into the verification pipeline

↓

Complete biometric verification

↓

Liveness check passed (system believes live person present)

↓

KYC completed (identity verified)

↓  
Account created OR existing account accessed

## Downstream payment consequence

The attack bypasses the core biometric trust mechanism. Once successful:

- Account created or accessed under a verified identity
- Payment access granted
- Fraudulent transactions authorized
- Funds moved or stolen
- Account takeover (if existing account) or synthetic account (if new)
- Mule activity or cash-out

## C. Detection Intelligence

| Signal Category      | Specific Signals                                                       | Detection Method           |
|----------------------|------------------------------------------------------------------------|----------------------------|
| Identity/KYC         | Biometric match anomalies                                              | ML                         |
|                      | Liveness check failures/retries                                        | Rule-based + ML            |
|                      | Inconsistent facial features (lighting, skin texture, eye movement)    | ML                         |
|                      | Synthetic artifact patterns in video                                   | ML                         |
|                      | Audio-visual desynchronization                                         | ML                         |
| Device/Session       | Virtual camera detection                                               | Rule-based + ML            |
|                      | Device emulation detection                                             | Rule-based                 |
|                      | Unusual camera pipeline behavior                                       | ML                         |
|                      | Non-rooted device with virtual camera app                              | ML                         |
| Behavioral           | Unnatural facial movements                                             | ML (Behavioral biometrics) |
|                      | Lack of appropriate micro-expressions                                  | ML                         |
|                      | Timing inconsistencies (response latency)                              | ML                         |
|                      | Unnatural speech patterns                                              | ML                         |
| Transaction          | (Post-account) transactions inconsistent with victim's normal behavior | ML                         |
| Network/Relationship | Multiple biometric attempts from same device                           | ML                         |

| Signal Category | Specific Signals                      | Detection Method             |
|-----------------|---------------------------------------|------------------------------|
| Temporal        | Same deepfake pattern across accounts | ML (Deepfake fingerprinting) |
|                 | Biometric attempts at unusual times   | Rule-based                   |
|                 | Rapid completion of verification      | ML                           |

## Evasion considerations

An adaptive attacker could:

- Use increasingly sophisticated injection techniques to evade hardware detection
- Improve deepfake quality to eliminate artifacts
- Add realistic micro-expressions and natural movements
- Rotate devices and virtual camera configurations
- Use unique deepfake patterns for each attempt

## D. Simulation Intelligence

| Field                           | Value                                                                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simulation type                 | Hybrid (Algorithmic + Agentic)                                                                                                                                                                        |
| Generation parameters           | Deepfake quality levels; Injection method variants; Target identity profiles; Verification flow interactions                                                                                          |
| Synthetic entities required     | Synthetic video streams (deepfake content) for injection; Synthetic audio (voice clone content); Device/session data showing injection attempt; Biometric verification flow interactions              |
| Synthetic transactions required | Transaction sequences following successful injection (may mimic victim's normal behavior or be fraudulent)                                                                                            |
| Temporal behavior               | Verification duration (abnormal if too fast); Response timing patterns; Liveness challenge responses                                                                                                  |
| Network relationships           | Same injection pattern across accounts; Same deepfake artifact signature; Connected to fraud rings                                                                                                    |
| Legitimate counterpart          | Users with poor camera quality or lighting; Users with physical characteristics that make liveness detection challenging; Users with special needs (RBI requires liveness checks not to exclude them) |
| Edge cases                      | Partial injection attempts (some checks pass, some fail); Users with varying camera quality; Users with disabilities affecting liveness response                                                      |

## E. Evidence

| Claim                                                                       | Source                             | Confidence |
|-----------------------------------------------------------------------------|------------------------------------|------------|
| Deepfakes account for 20% of biometric fraud attempts                       | Entrust 2026 Identity Fraud Report | Verified   |
| Deepfake selfies increased 58% in 2025                                      | Entrust 2026 Identity Fraud Report | Verified   |
| Injection attacks increased 40% year-over-year                              | Entrust 2026 Identity Fraud Report | Verified   |
| Crypto targeted in 60% of deepfake cases                                    | Entrust 2026 Identity Fraud Report | Verified   |
| MITRE ATLAS published iProov deepfake KYC injection case study              | MITRE ATLAS / iProov               | Verified   |
| Attack uses Faceswap + OBS + Virtual Camera on non-rooted Android           | iProov/MITRE ATLAS                 | Verified   |
| FATF Horizon Scan on AI and Deepfakes published 22 December 2025            | FATF                               | Verified   |
| FATF named liveness check as attack surface                                 | FATF Horizon Scan                  | Verified   |
| RBI August 2025 amendment mandated active deepfake/spoof detection in V-CIP | RBI KYC Master Direction amendment | Verified   |

## F. Product Mapping

| Component            | Implementation                                                                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generate module      | Agentic generation of deepfake video/audio content; Algorithmic generation of device/session data; Hybrid generation of verification flow interactions |
| Defend module        | AI-driven deepfake detection; Hardware-level sensor validation; Behavioral biometrics                                                                  |
| Detection model      | Deepfake artifact detection (CNN/Transformer); Liveness detection with multi-factor verification; Behavioral biometrics for facial movements           |
| Potential mitigation | Multi-layer verification (hardware-based + hybrid liveness); AI-driven content validation tools; Human expertise for contextual anomalies              |
| Evaluation metric    | False Acceptance Rate (FAR); False Rejection Rate (FRR); Speed (real-time verification required)                                                       |

# FAMILY 4: GENAI-POWERED SOCIAL ENGINEERING / PERSON-NOT-PRESENT (PNP)

## A. Identity

| Field                          | Value                                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Attack ID                      | SEP-001                                                                                                                   |
| Attack Family                  | GenAI-Powered Social Engineering / Person-Not-Present (PNP)                                                               |
| Attack Subtypes / Variants     | Phishing (email); Vishing (voice); Romance scams; Digital Arrest scams; Business Email Compromise (BEC); Investment scams |
| Primary Lifecycle Stage        | Identity / KYC (Stage 1) – post-verification exploitation                                                                 |
| Secondary / Cross-stage Stages | Payment initiation (Stage 5); Beneficiary / Payee (Stage 12); Cash-out / Mule (Stage 16)                                  |
| Target / Actor                 | Legitimate KYC-verified customers, employees, merchants                                                                   |
| Attack Objective               | Manipulate a legitimate user into authorizing a fraudulent payment or revealing credentials                               |

## B. Fraud Understanding

### What is the attack?

Use of GenAI to create personalized, scalable social engineering attacks that manipulate legitimate users into authorizing fraudulent payments. The FATF notes "criminals are increasingly exploiting generative AI, AI agents, and deepfake technologies to enable fraud, scale illicit finance, and undermine trust in financial systems".

### Traditional mechanism

Before GenAI, social engineering relied on phishing emails (often with poor grammar and generic content), phone scams (human callers with scripts), romance scams (manual engagement), and Business Email Compromise (email account compromise). These required social engineering skills, script-writing ability, and patience for manual engagement.



# GenAI transformation

GenAI enables **industrialized, personalized social engineering**:

- Criminals can analyse large volumes of publicly available data and craft highly targeted behavioural scams
- With affordable GenAI tools, fraudsters can create convincing phishing messages, synthetic voice clones, and deepfake videos at scale
- Personalized scripts adapted to each victim in real-time

## Why GenAI is load-bearing

If GenAI is removed, the attacker loses:

- Ability to generate personalized phishing at scale
- Ability to adapt scripts in real-time
- Ability to create convincing synthetic voices for call scams
- Ability to automate victim engagement

The attack shifts from **mass, generic** social engineering to **targeted, personalized, scalable** campaigns.

## Prerequisites

- Access to GenAI tools (LLMs for script generation, voice cloning, deepfake video)
- Target intelligence (social media, public data)
- Communication channels (email, phone, SMS, video, social media)

## Attack flow

Gather intelligence on target (social media, public data)

↓

Generate personalized scam script using GenAI

↓

Execute via appropriate channel (voice, email, video, messaging)

↓

Manipulate victim into authorizing payment

↓

Victim authorizes payment voluntarily (appears legitimate)

↓

Funds transferred to mule accounts

↓

Layering across multiple accounts

↓  
Cash-out

## Downstream payment consequence

**Key insight:** The user completes the transaction voluntarily. Rule-based defenses fail because the transaction appears legitimate. This is the defining characteristic of PNP fraud—the fraud occurs after KYC, exploiting the trust established during onboarding.

Real-world examples:

- **"Digital Arrest" scams in India:** Losses exceeded ₹120 crore in just one quarter of 2024. Attackers impersonate law enforcement via deepfake calls.
- **Romance scams with AI personas:** Increased significantly in 2025.
- **Automated scam campaigns:** Semi-automated scams using ChatGPT-generated content have been documented.

## C. Detection Intelligence

| Signal Category | Specific Signals                                                               | Detection Method |
|-----------------|--------------------------------------------------------------------------------|------------------|
| Identity/KYC    | (Victim is legitimate—no KYC signals)                                          | N/A              |
| Device/Session  | Unusual login location/time for victim                                         | Rule-based + ML  |
|                 | New device detection                                                           | Rule-based       |
|                 | Session anomalies                                                              | ML               |
| Behavioral      | Unusual transaction behavior (new beneficiary, unusual amount, unusual timing) | ML               |
|                 | Signs of coercion (rapid authentication, unusual navigation)                   | ML               |
|                 | Unusual interaction patterns                                                   | ML               |
| Transaction     | New beneficiary addition                                                       | Rule-based + ML  |
|                 | Unusual transaction amount (relative to victim's normal behavior)              | ML               |
|                 | Unusual transaction velocity                                                   | ML               |

| Signal Category      | Specific Signals                                      | Detection Method     |
|----------------------|-------------------------------------------------------|----------------------|
|                      | Transaction pattern inconsistent with normal behavior | ML                   |
|                      | Rapid transaction sequences                           | ML                   |
| Network/Relationship | Beneficiary connected to known mule networks          | ML (Graph analytics) |
|                      | Funds move to suspicious accounts                     | ML                   |
|                      | Beneficiary risk score                                | ML                   |
| Temporal             | Transaction at unusual time                           | Rule-based + ML      |
|                      | Time between scam contact and payment                 | ML                   |
|                      | Rapid sequence of transactions                        | ML                   |

## Evasion considerations

An adaptive attacker could:

- Make transactions appear more normal (amounts, timing, frequencies)
- Use varied beneficiary accounts with no obvious connections
- Mimic victim's normal transaction patterns
- Use victim's own device (if compromised) or mimic victim's device fingerprint
- Create complex layering with no obvious network patterns

## D. Simulation Intelligence

| Field                           | Value                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simulation type                 | Agentic (primary) with Algorithmic (transaction component)                                                                                                                                                     |
| Generation parameters           | Scam type distribution; Victim persona profiles; Communication channel selection; Script generation parameters; Transaction sequence generation                                                                |
| Synthetic entities required     | Victim personas (legitimate KYC-verified users with transaction histories); Attacker personas; Trusted authority personas (if impersonation); Scam scenarios (phishing, vishing, romance, digital arrest, BEC) |
| Synthetic transactions required | New beneficiary addition; Fraudulent payment authorization; Funds movement to mule accounts; Layering transactions                                                                                             |

| Field                  | Value                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temporal behavior      | Time between scam contact and payment (hours to days); Time of day patterns; Transaction velocity patterns                                                            |
| Network relationships  | Beneficiary connected to mule network; Funds layering path; Connections to fraud rings                                                                                |
| Legitimate counterpart | Legitimate new beneficiary payments (e.g., paying a new vendor); Legitimate unusual transactions (e.g., emergency payments); Users with variable transaction patterns |
| Edge cases             | Partial scams (victim resists, partial payment); Multiple communication channels; Different scam types (romance, investment, impersonation, digital arrest)           |

## E. Evidence

| Claim                                                        | Source                              | Confidence           |
|--------------------------------------------------------------|-------------------------------------|----------------------|
| FATF warns criminals exploiting GenAI and deepfakes          | FATF Horizon Scan                   | Verified             |
| "Digital Arrest" scams: Indians lost ₹120.3 crore in Q1 2024 | Government of India cybercrime data | Verified             |
| RBI flags social engineering powered by GenAI/deepfakes      | RBI                                 | Verified             |
| GenAI enables personalized, scalable social engineering      | Industry analysis                   | Reasonable inference |

## F. Product Mapping

| Component            | Implementation                                                                                                                                                        |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generate module      | Agentic generation of scam scenarios, dialogue, and messages; Algorithmic generation of transaction sequences; Synthetic persona generation for victims and attackers |
| Defend module        | Behavioral analytics for transaction anomalies; Graph analytics for beneficiary network analysis; Real-time anomaly detection                                         |
| Detection model      | Behavioral anomaly detection; Graph Neural Network for beneficiary risk scoring; Temporal pattern analysis                                                            |
| Potential mitigation | Real-time transaction monitoring; Beneficiary risk scoring; Customer education and alerts; Delayed settlement for high-risk transactions                              |
| Evaluation metric    | Detection rate for scam-induced payments; False positive rate on legitimate new beneficiary payments; Speed (real-time detection needed for payment blocking)         |

# FAMILY 5: ACCOUNT TAKEOVER VIA KYC RECOVERY EXPLOITATION

## A. Identity

| Field                          | Value                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Attack ID                      | ATO-001                                                                                                               |
| Attack Family                  | Account Takeover via KYC Recovery Exploitation                                                                        |
| Attack Subtypes / Variants     | Password reset impersonation; Security question bypass; Customer support impersonation; Document-based recovery fraud |
| Primary Lifecycle Stage        | Identity / KYC (Stage 1) – recovery verification                                                                      |
| Secondary / Cross-stage Stages | Authentication (Stage 4); Account takeover                                                                            |
| Target / Actor                 | KYC recovery processes, password reset flows, customer support                                                        |
| Attack Objective               | Impersonate legitimate account holders during KYC recovery processes to gain account access                           |

## B. Fraud Understanding

### What is the attack?

Use of GenAI-generated documents and deepfakes to impersonate legitimate account holders during KYC recovery processes (password reset, security questions, customer support verification), enabling account takeover.

### Traditional mechanism

Before GenAI, account takeover via recovery relied on:

- Stolen credentials (username/password from data breaches)
- Social engineering of support staff

- Guessing security questions
- SIM swapping

## GenAI transformation

GenAI enables attackers to generate convincing impersonation evidence on demand:

- Forged identity documents matching the victim's profile
- Voice clones to pass phone-based verification
- Deepfake video to pass video-based recovery verification

## Why GenAI is load-bearing

If GenAI is removed, the attacker loses:

- Ability to generate convincing impersonation evidence without physical access to victim documents
- Ability to create voice clones on demand
- Ability to produce video evidence for video-based recovery

## Prerequisites

- Victim's PII (from data breaches)
- Access to GenAI tools (document generation, voice cloning, deepfake video)
- Knowledge of recovery verification process

## Attack flow

Obtain victim's PII (data breach, social media, phishing)

↓

Generate forged identity documents using GenAI

↓

Create voice clone or deepfake of victim (if needed)

↓

Initiate account recovery process

↓

Submit forged documents / pass voice verification

↓

Recovery KYC bypassed

↓

Account access granted

↓  
Account takeover

## Downstream payment consequence

Account takeover

↓

Access to victim's financial accounts

↓

Funds drained

↓

New beneficiaries added

↓

Fraudulent transactions authorized

↓

Funds transferred to mule accounts

↓

Layering and cash-out

---

## C. Detection Intelligence

| Signal Category      | Specific Signals                                    | Detection Method |
|----------------------|-----------------------------------------------------|------------------|
| Identity/KYC         | Document anomalies in recovery submissions          | ML               |
|                      | Biometric match anomalies                           | ML               |
|                      | Inconsistent identity data                          | Rule-based + ML  |
| Device/Session       | Unusual device for recovery request                 | Rule-based       |
|                      | Unusual location/IP                                 | Rule-based + ML  |
|                      | Multiple recovery attempts                          | Rule-based       |
| Behavioral           | Unusual interaction patterns                        | ML               |
|                      | Signs of automated behaviour                        | ML               |
| Transaction          | Unauthorized transactions post-recovery             | ML               |
|                      | New beneficiary additions                           | ML               |
|                      | Funds movement to mule accounts                     | ML               |
| Network/Relationship | Recovery attempts connected to known fraud patterns | ML               |

| Signal Category | Specific Signals          | Detection Method |
|-----------------|---------------------------|------------------|
| Temporal        | Recovery at unusual times | Rule-based       |
|                 | Rapid recovery completion | ML               |

## Evasion considerations

An adaptive attacker could:

- Use high-quality forged documents that pass verification
- Use realistic voice clones for phone verification
- Mimic victim's normal behaviour post-recovery
- Use victim's own device (if compromised)

## D. Simulation Intelligence

| Field                           | Value                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simulation type                 | Agentic                                                                                                                                               |
| Generation parameters           | Recovery channel selection; Document forgery parameters; Voice clone quality; Interaction sequences                                                   |
| Synthetic entities required     | Victim personas (legitimate account holders with transaction histories); Forged identity documents; Voice clone audio; Recovery interaction sequences |
| Synthetic transactions required | Post-recovery unauthorized transactions; New beneficiary additions; Funds movement to mule accounts                                                   |
| Temporal behavior               | Recovery timing (off-hours weighted); Post-recovery transaction patterns                                                                              |
| Network relationships           | Recovery attempts connected to known fraud patterns; Mule network connections                                                                         |
| Legitimate counterpart          | Legitimate account recovery (lost password, forgotten security questions); Legitimate customer support interactions                                   |
| Edge cases                      | Partial recovery attempts; Multiple recovery channels; Different recovery verification requirements                                                   |

## E. Evidence



| Claim                                                                            | Source                             | Confidence           |
|----------------------------------------------------------------------------------|------------------------------------|----------------------|
| Account takeovers and sophisticated synthetic identities can pass initial checks | PYMNTS/Mastercard/Trulioo report   | Verified             |
| Fraudsters use GenAI to create convincing impersonation evidence                 | Industry analysis                  | Reasonable inference |
| Deepfakes used to bypass authentication                                          | Entrust 2026 Identity Fraud Report | Verified             |

## F. Product Mapping

| Component                   | Implementation                                                                                                                                       |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generate module</b>      | Agentic generation of recovery scenarios; Algorithmic generation of forged documents and transaction sequences                                       |
| <b>Defend module</b>        | Multi-factor verification for recovery; Document forensics; Voice biometrics; Behavioral analytics                                                   |
| <b>Detection model</b>      | Anomaly detection for recovery attempts; Document forgery detection; Voice biometric matching                                                        |
| <b>Potential mitigation</b> | Enhanced verification for recovery (additional factors); Delayed recovery processing for high-risk cases; Customer notification of recovery attempts |
| <b>Evaluation metric</b>    | False Acceptance Rate (FAR) for recovery attempts; False Rejection Rate (FRR) for legitimate recoveries; Speed                                       |

# Summary: Final Identity/KYC Attack-Family Taxonomy

| # | Attack Family                      | GenAI Load-Bearing                | Simulation Type | Primary Stage |
|---|------------------------------------|-----------------------------------|-----------------|---------------|
| 1 | Synthetic Identity Fraud           | Yes – industrial-scale generation | Algorithmic     | Identity/KYC  |
| 2 | GenAI Document Forgery             | Yes – off-the-shelf capability    | Algorithmic     | Identity/KYC  |
| 3 | Deepfake Injection / Impersonation | Yes – qualitatively new vector    | Hybrid          | Identity/KYC  |

| # | Attack Family                          | GenAI Load-Bearing            | Simulation Type | Primary Stage                      |
|---|----------------------------------------|-------------------------------|-----------------|------------------------------------|
| 4 | GenAI-Powered Social Engineering / PNP | Yes – personalized, scalable  | Agentic         | Identity/KYC (post-verification)   |
| 5 | Account Takeover via KYC Recovery      | Yes – on-demand impersonation | Agentic         | Identity/KYC $\cap$ Authentication |

**Total: 5 distinct attack families**